Asynchronous YOSO à la Paillier

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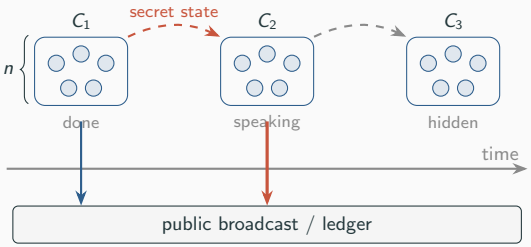
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YOSO: You Speak Only Once

- Like Nakamoto: a party wins a secret lottery, sends one block, and is done
- YOSO: a secret committee sends one message each and hands its secret state to the next
- The adversary may corrupt anyone, but only after they have spoken
- Only n of the M parties speak per step, but nobody knows who is next, so every message goes to all M



population of $M \gg n$ parties; committees are sampled from it

The setting of earlier YOSO MPC

	Network	Broadcast	Roles
YOSO model [GHK+21]	sync	assumed	ideal
YOSO MPC [KRY22,BDO23]	sync	assumed	ideal
Fluid / layered MPC	sync	assumed	known
Role assignment [BGG+20,GHM+21,CKR+23]	sync	assumed	**built**
This paper	**async**	**built** (TOB)	**built**

Our approach

Asynchrony

No timeouts: every step waits for $n - t$ of a committee, and any committee can be late.

Broadcast

YOSO total-order broadcast: reliable broadcast + agreement on a core set, with self-nominated committees.

Coins come from the MPC.

Role assignment

Anonymous roles and secret channels to them, from threshold Paillier: PIR + one-time pads.

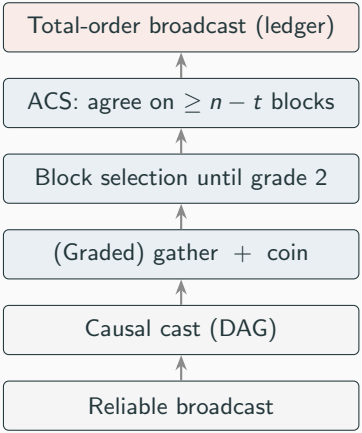
Adaptive, $(\frac{1}{3} - c)$, no FHE, no NCE.

Secure against $T < (\frac{1}{3} - c)M$ adaptive corruptions, in the random-oracle model with a one-time setup

Broadcast to future committees

- YOSO, fluid and layered MPC assume a broadcast channel to the next committee
- In the *layered* model: deterministic broadcast is impossible [Gar94,DDG+23], and randomized synchronous broadcast was characterized recently [DLM+26]
- We need it asynchronous and totally ordered, with each role speaking once, to parties nobody can name yet
- Each committee message goes to all M parties

Total-order broadcast: the stack

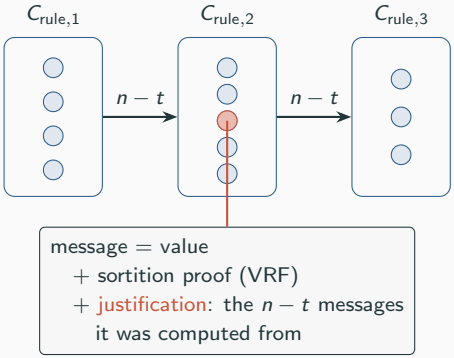


Starting point: asynchronous TOB from reliable broadcast + a coin [KN23]

Making it YOSO:

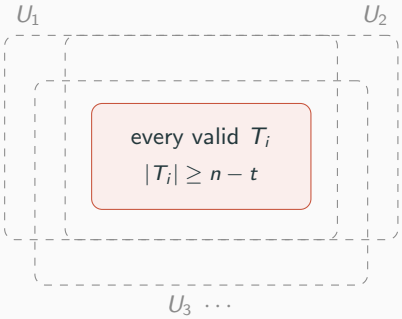
1. every activation rule is run by a fresh committee
2. RB delivers to all M parties: the next committee is unknown
3. leader election without a committee list

One committee per activation rule



- Self-nominate per (session, rule) by VRF sortition [GHM+17]: unpredictable, verifiable, speak once
- No role assignment needed: TOB never sends secrets to the future
- Whp each committee has enough honest members (a majority, or two thirds for RB), but unknown members and size

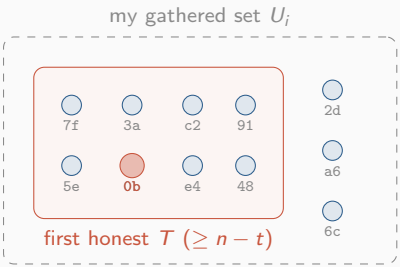
Block selection with a known committee [KN23]



king $S_k \in T_i$	grade 2
king $S_k \in U_i \setminus T_i$	grade 1
otherwise	grade 0 (keep my block)

1. Everyone inputs a block. Graded gather gives party i sets $T_i \subseteq U_i$
2. Guarantee: every valid T_i is inside every valid U_j
3. *Then* a coin names a king S_k, $k \in [n]$. Output its block with a grade
4. Grade 2 anywhere $\Rightarrow$ king in every U_j $\Rightarrow$ everyone has grade ≥ 1, same block
5. The first honest T_i is fixed before the coin and covers $> n/2$: grade 2 with prob. $\geq \frac{1}{2}$. Repeat until grade 2

Leader election without a committee list



$\text{ticket}_j = H(P_j \,\|\, \text{coin})$, candidate = smallest ticket

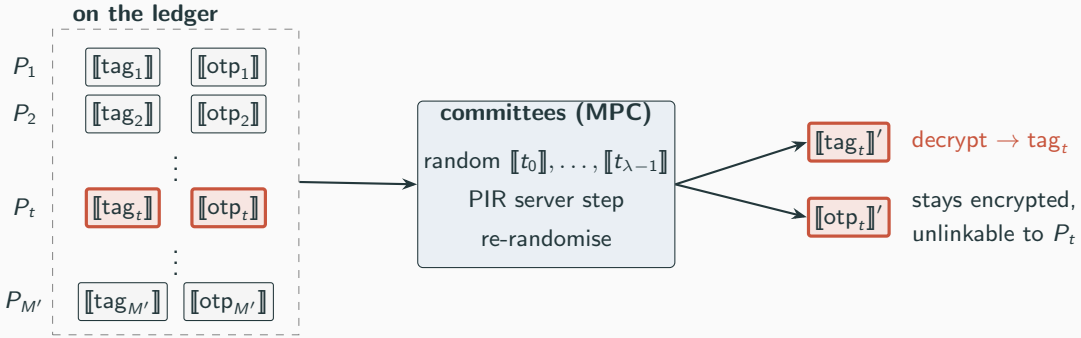
- With sortition there are no indices and no known committee size, so the coin can't name a king
- Instead: rank the members *I have seen* (U_i) by a hash of the coin. My candidate has the smallest ticket
- Coin flipped after the first honest T is fixed: the global winner is in it with constant probability, hence in every U_j
- Two vote rounds turn candidates into grades $0/1/2$, as before

The missing coin: where the MPC comes in

- Every round of block selection needs a coin that nobody can predict
- The coin comes from the MPC: it shares a random value onto a committee, which opens it when asked
- The MPC follows CDN [CDN01]: state is public Paillier ciphertexts on the ledger, $+$ is local, $\times$ and output need the committee's key shares
- The only secret is the decryption key, held in shares by a committee that must hand it on to the next

$\Rightarrow$ That next committee must be secret, so the MPC needs role assignment

Role assignment = a PIR client, run inside MPC



- New 2-message PIR from Damgård–Jurik: the client only holds Paillier ciphertexts, so the MPC can be the client
- P_t recognises its tag, and nobody else learns who holds the role
- Asynchrony: the database is whichever $M - T$ inputs arrive first

A 2-message PIR from Damgård–Jurik

Database: $y_0, \ldots, y_{L-1}$, with $L = 2^\lambda$

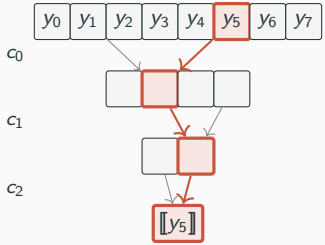
Client: for each bit t_i of the index, sends $c_i = \mathsf{E}_{i+1}(t_i)$ at Damgård–Jurik level $i{+}1$

Server: with $\bar{c}_i$ an encryption of $1 - t_i$, halve the list once per bit: $y_j^{i+1} \leftarrow c_i^{y_j^i} \cdot \bar{c}_i^{y^i_{j+L_i/2}}$

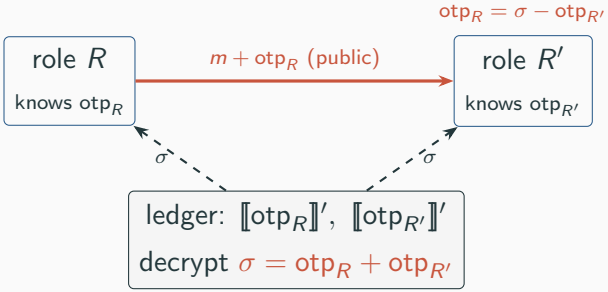
Why it works: level-s plaintexts are level-$(s{-}1)$ ciphertexts

Answer: y_t under one layer of encryption per index bit

In MPC: the index bits are random encrypted bits, anyone can compute the server step, and the committees re-randomise and decrypt the outer layers, leaving a fresh encryption of y_t



Secret channels to unknown parties: no role keys



- Earlier work: an anonymous public key per role
- Here: a decrypted sum is a shared one-time pad between two parties who don't know each other
- Pads are committed, so resharing of the key is verifiable

The price: circularity

Key shares travel under pads that are encrypted under the same Paillier key.

The price: a circular selective-opening assumption

- Key shares of d travel under pads, and the pads are encrypted under the same key
- CSO game: a box holds secret state derived from d
- The adversary sees encryptions of functions of that state, plus admissible leakage
- It must tell whether a public element w_S encrypts 0 or 1
- Without d in the state, this follows from CPA. We assume the link to d does not help
- Our protocol only needs one box: it reshares d, encrypts the shares, and leaks unqualified sets of shares

Passing the key on, adaptively

- d is shared over a *pair* of committees: additively, then each additive share with threshold $t < n/2$
- Linear integer secret sharing [DT06] instead of integer Shamir: share size stays constant over time
- Each step reshares to the next pair, over the pad channels. Pads are committed, so resharing is verifiable
- Adaptive security in the style of [DN03]. Its simulator trapdoor doesn't fit YOSO, so we take one from the random oracle
- Single-inconsistent-party technique, adapted to asynchronous YOSO

Takeaways

- Asynchronous YOSO MPC with broadcast and role assignment built in, from black-box threshold Paillier:
  - YOSO total-order broadcast: a fresh committee per step, leader election without a committee list
  - PIR inside MPC $\Rightarrow$ anonymous roles, and summed one-time pads $\Rightarrow$ channels to them

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